AMT Worksheet 13
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The purpose of this assignment is to illustrate the use of Gaussian elimination to solve systems of equations, the meaning of linear independence, and the behavior of matrices acting on various vector spaces. All vector spaces in this assignment are understood in the framework of the real number system, with ordinary addition and scalar multiplication operations.

In problems 1 – 4, determine whether or not the given set constitutes a vector space. Explain your reasoning.

1. The set of polynomials of degree 6.

2. The set of polynomials of degree 2 or lower.

3. The set of vectors in $\mathbb{R}^3$ of the form $\begin{pmatrix} a+2b \\ 2b+1 \\ a-b \end{pmatrix}$, where a and b are real numbers.

4. The set of vectors in $\mathbb{R}^3$ of the form $\begin{pmatrix} 4a-2b+2 \\ 2a-b+1 \\ -2a-2b \end{pmatrix}$, where a and b are real numbers.

5. Show that the distributive property implies that $0|x\rangle = 0$ for every vector in a given space.

6. Use Gaussian elimination to find the most general solution to the system $\begin{aligned} 3x-2y+2z &= 8 \\ -x+2y-z &= -3 \\ x+6y-z &= 1 \end{aligned}$.

7. Use a cofactor expansion to determine the inverse of the matrix $\begin{pmatrix} 1 & 1 & 0 \\ 2 & -1 & 2 \\ -1 & 3 & 1 \end{pmatrix}$. Compare your inverse to that given by your calculator.

8. Find the determinant of the matrix $\begin{pmatrix} 3 & 2 & -2 & -4 \\ 2 & 1 & 1 & 3 \\ -1 & 3 & 7 & 2 \\ 4 & 3 & 3 & 1 \end{pmatrix}$ *without a calculator*. Your work should indicate that you did not require a calculator….

9. Determine the number of pivots in the matrix $M = \begin{pmatrix} 2 & 3 & -5 & 2 \\ -1 & 1 & -5 & -3 \\ 3 & 2 & 2 & 5 \end{pmatrix}$. What is the rank of this matrix? Determine the dimension of the row space and column space of this matrix. Does this matrix map onto, one-to-one, or both when considered a mapping from $\mathbb{R}^4$ to $\mathbb{R}^3$? Find a basis for the null space of this matrix. What is the dimension of this space? Are the columns of this matrix linearly independent? Are the rows? Explain. Is it possible for the equation $M|x\rangle = |y\rangle$ to have no solutions? Is it possible to have infinitely many solutions? Is it possible for this equation to have exactly one solution? Explain.

10. Determine the number of pivots in the matrix $\begin{pmatrix} 2 & 5 \\ 1 & 3 \\ -2 & 4 \end{pmatrix}$. What is the rank of this matrix? Determine the dimension of the row space and column space of this matrix. Does this matrix map onto, one-to-one, or both when considered a mapping from $\mathbb{R}^2$ to $\mathbb{R}^3$? Find a basis for the null space of this matrix. What is the dimension of this space? Are the columns of this matrix linearly independent? Are the rows? Explain. Is it possible for the equation $M|x\rangle = |y\rangle$ to have no solutions? Is it possible to have infinitely many solutions? Is it possible for this equation to have exactly one solution? Explain.

11. Write the vector $|x\rangle = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ in terms of the basis $B = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \end{pmatrix} \right\}$.

12. Find a set of matrices that map between $\mathbb{R}^2$ and the space $V = \text{span}\left\{ \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \right\}$ in a one-to-one and onto manner (so one matrix for $\mathbb{R}^2$ to V and another for V to $\mathbb{R}^2$). Explain why you are guaranteed to be able to find a set of matrices that accomplish this transformation and why the answers to this question are not unique. Determine the product of the two matrices you found in *both* orders, and explain the relationship these products have with the identity matrix. Is it possible for both products to equal the identity matrix? Explain.

13. Find the eigenvectors and eigenvalues of the following matrices. In each case, state whether or not the eigenvectors span the space on which the matrix acts. If they *do* span the space, determine the matrices that play the role of the square root of the matrix. You may use your calculator to check the answers you obtain, but your work should demonstrate that a calculator is not necessary.

(a) $\begin{pmatrix} 15 & -11 \\ 6 & -2 \end{pmatrix}$ (b) $\begin{pmatrix} 7 & 0 & -2 \\ 0 & 3 & -11 \\ 4 & 0 & 1 \end{pmatrix}$ (c) $\begin{pmatrix} 13 & 1 & 6 \\ 0 & 15 & 0 \\ 6 & 0 & -3 \end{pmatrix}$

14. Find the eigenvalues and eigenvectors of the given matrix, then determine the given function of that matrix.

(a) $A = \begin{pmatrix} 7 & -4 \\ 6 & -3 \end{pmatrix}$. Find 2^A. (b) $A = \begin{pmatrix} 11 & 25 \\ 2 & 6 \end{pmatrix}$. Find $\log_2 A$.

15. Is it possible for the dimension of the row space of a matrix to differ from that of the column space? Explain.

16. A matrix of rank 3 has 15 columns and 6 rows. Determine the dimension of the row space, column space, and null space of this matrix. How many free parameters are there in the solution of a linear system with this matrix as the coefficient matrix, assuming that there are solutions? Is it possible for there to be no solutions to this system? Is it possible for this system to have exactly one solution? Explain.